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EFFECTS OF FISH OIL AND LIPID LOWERING DRUGS ON SERUM TRIGLYCERIDES ALONE AND IN COMBINATION

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ABSTRACT

Antihypertriglyceridemic effects of fish oil alone and in combination with other lipid lowering drugs as lovastatin and gemfibrozil were discussed especially with reference to their effects on serum triglycerides which are found to be the major cause of atherosclerosis and ischemic heart disease. This paper describes the effects of fish oil in lowering plasma total fats and effects of lovastatin and gemfibrozil alone and in combination.

INTRODUCTION

Fish oil and fish may be useful components of diet for the treatment of hypertriglyceridemia. This effect is due to the presence of PUFAs (polyunsatutated fatty acids), ω-3 fatty acids, EPA (ecosapentaenoic acid) and DHA (decosahexaenoic acid) in fish oil. Many factors are found to be involved in causing plasma hypertriglyceridemia such as high fat diet, lipoprotein, personality, trait and genetic background of individuals (Castelli, 1984). The fish cause decrease in plasma triglyceride levels due to marked reduction in VLDL (very low density lipoprotein) triglyceride synthesis (Daggy *et al.,* 1987 and Abraham *et al.*, 1992). It should be noted that not only high consumption of fat and cholesterol in plasma leads to cardiac problems but their utilization is also necessary. Hyperlipoproteinemia is a pathological condition where fish and fish oil or dietary measures are useless in lowering TGAs (triglycerides), cholesterol and lipoproteins and the only source to treat the symptoms are the drugs such as lovastatin, gemfibrozil, clofibrate, cholestyramine, nicotinic acid, colestipol. Among these drugs lovastatin and gemfibrozil produce significant effects although this combination should be used cautiously because of the increased frequency of myositic complication (Tobert, 1988).

Fredrikson *et al.* (1967) describes five types of hyperlipoproteinemia. Type I or hyperchylomicronemia, the only type which does not require drug and can be controlled by diet therapy, caused due to the deficiency of lipoprotein lipase in serum, rest of the types require drug for their treatment. Type IIA hyper-betalipoproteinemia where plasma cholesterol levels are elevated but serum triglycerides remain within normal limits due to the deficiency of LDL (low density lipoprotein) receptors which increases LDL but not VLDL in plasma, whereas IIB type mixed hyperbetalipoproteinemia suffering patients show both elevated levels of LDL and VLDL and modest increase in TGAs in serum. Most authorities would agree that LDL cholesterol value of 5 mmol/l or above is associated with increased risk of atherosclerosis (Thompson G.R., 1989). In type III hyper-remnant lipoproteinemia, there is a deficiency of apoprotein E_3 in serum resulting elevating levels of TGs. Type IV hyper-betalipoproteinemia which is caused due to defective carbohydrate (CHO) metabolism again resulting in high levels of (VLDL) pre-betalipoproteinemia and triglycerides (TGs).

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Type V mixed lipidemia, a very rare disorder where LDL is reduced resulting excessive prebetalipo-proteinemia (VLDL) and chylomicrons. This paper studied the effects of fish oil and lovastatin on TGs levels in different types of hyperlipoproteinemias in plasma serum. Much stress is laid on the study of fish oils having high potentials of activities as hypolipidemic agents (Castro *et al.,* 1997, Anil *et al.*, Van Vilijman *et al.*, 1998).

MATERIALS AND METHODS

Electrical centrifuge 1000-4000 rpm (China)
Disposable syringe 10 ml, 21 gauge (B.P.)
Centrifuge tube
Glass tube
Pasture pipettes
Juster 0.1 ml to 1 ml
Microlab 100 (E. Merck)

Reagents:
All the reagent kits were purchased from Federal Republic of Germany.

Experimentally prepared food materials:
Cholesterol (Cruda, USA)
Butter (Lurpak, Denmark)
Lovastatin
Gemfibrozil
Fish Oil
Chick pea (gram seed), local supplier
Alfa alfa (loosen or green grass), local supplier

Animals and their feeding schedule
All the procedure is described in detail elsewhere in the thesis (Faheem, 1995).

RESULTS AND DISCUSSION

Results of fish oil and lipid lowering drugs on serum triglycerides alone and in combination are shown in Tables 1-9.

During the course of experiments where fish oil is given along with gemfibrozil and lovastatin, overall effect was observed to be 74.25% ($P<0.01$) which is lower than the combination of fish oil + gemfibrozil which is 79.58% ($P<0.01$) and a better profile than that of fish oil + lovastatin i.e., 63.13% ($P<0.01$). During the same experiment, lovastatin alone reduced the TGs level upto 57.14% ($P<0.05$), whereas gemfibrozil showed a decrease of 48.02% ($P<0.01$). However the combination of gemfibrozil + lovastatin exhibited the best result among all the combinations and decreased the triglyceride level to a maximum value of 84.06% ($P<0.05$).

These significant results can be explained by the different properties of the two drugs. The effect of HMG-CoA reductase inhibitors are mediated by way of increased LDL receptor-mediated removal of LDL-cholesterol from the circulation as well as from decrease LDL production (Brown and Goldstein, 1986; Grundy, 1988).

Fibrates mainly act by increasing lipolytic degradation of VLDL and by reducing VLDL synthesis and secretion, and therefore have only limited LDL cholesterol reducing properties (Grundy and Vega, 1987). The role of gemfibrozil as a potential therapeutic agent for hypertriglyceridemia is well established.

Several explanations for the triglyceride lowering effect of gemfibrozil exist but the lesser effect brought about by lovastatin has not been clarified. This effect is not due to lipolytic activity because no changes in postheparin plasma lipase activities were noticed during lovastatin treatment (Helve and Tikkansen, 1988). It has also been proposed that cholesterol synthesis inhibition by lovastatin could be linked to inhibition of synthesis of apolipoprotein β-containing particle (Ginsberg et al., 1987), thus leading to reduced hepatic output of triglycerides rich lipoprotein particle.

The best and significant effects on triglycerides are of the combination of gemfibrizil and lovastatin. Dietary fish oil has been shown to block the usual induction of hypertriglyceridemia, due to an abrupt increase in CHO (Harris et al., 1984). This group of workers also investigated (1990) that ω-3 fatty acids of fish oil markedly lowered plasma triglycerides and very low density lipoprotein (VLDL) levels in both normal and hypertriglyceridemic subjects. The hypertriglyceridemic effect of fish oil caused primarily by an inhibition of VLDL triglyceride synthesis, but an additional, independent effect upon VLDL catabolism cannot be ruled out.

Table-1

Effect Of Normal Diet On the Serum Triglycerides mg/100 ml in Group-1 Normal Control (N.C) Rabbits

Weeks	1A	1B	1C	1D	1E	MEAN	± SE
0	47	62	84	65	56	58.80	2.99
4	64	85	65	70	63	65.40	1.08
8	70	77	80	76	74	75.40	1.49
12	85	89	85	89	89	87.40	0.08
18	90	99	98	95	93	95.00	1.47
18	99	108	105	102	108	104.40	1.56
20	105	114	113	112	114	111.60	1.51
22	112	122	125	123	129	122.20	2.52

Table-2

Effect of Cholesterol + Butter Rich Diet on the Serum Triglycerides mg/100 ml in Group-2 Pathological Control (P.C) Rabbits

Weeks	2A	2B	2C	2D	2E	MEAN	±SE
0	22	4.8	59	45	58	46.40	5.98
4	89	177	212	181	96	151.00	22.05
8	150	231	401	245	132	233.00	42.19
U	187	353	514	363	203	324.00	53.62
14	210	425	600	475	295	401.00	61.11
Effect of normal diet*							
18	190	380	510	375	235	338.00	51.06
20	186	286	436	281	192	276.00	40.53
22	148	22.0	375	225	122	218.00	39.40

*Normal diet Loosan (alfa alfa), Gram seeds (chick pea)

Table-3
Effect of Cholesterol + Butter Rich Diet on the Serum Triglycerides mg/100 ml
in Group 3 Treated 1 (T.1) Rabbits

Weeks	3A	38	3C	3D	3E	MEAN	*SE
0	59	25	28	67	87	49.20	8.40
4	80	58	55	78	108	75.80	8.51
8	93	81	68	97	115	90.80	7.06
12	102	97	95	115	131	108.00	8.01
16	120	140	119	130	140	129.80	410
Effect of fish oil (71.43 mg/kg/day)							
18	112	129	105	120	75	108.20	8.24
20	104	110	85	68	50	83.40	9.97
22	60	75	82	50	30	59.40	8.26

Table-4
Effect of Cholesterol + Butter Rich Diet on the Serum Triglycerides mg/100 ml
in Group-4 Treated 2 (T.2) Rabbits

Weeks	4A	4B	4C	4D	4E	Mean	± SE
0	21	15	36	64	23	31.80	7.83
4	25	28	43	69	27	38.00	7.54
8	29	33	63	72	35	46.40	7.86
12	41	39	66	83	46	55.00	7.58
18	68	42	74	88	57	65.80	6.95
Effect of Lovastatin (0.57 mg/kg/day)							
18	49	37	58	67	49	52.00	4.49
20	26	35	46	57	36	40.00	4.74
22	18	31	19	50	23	28.20	5.29

Table-5
Effect of Cholesterol + Butter Rich Diet on the Serum Triglycerides mg/100 ml
in Group-6 Treated 3 (T.3) Rabbits

Weeks	5A	SB	SC	SD	SE	Mean	*SE
0	33	10	7	5	11	13.20	4.53
4	61	31	17	38	45	38.40	6.53
8	76	39	28	71	75	57.80	9.04
12	90	56	46	87	90	73.80	8.46
16	96	85	67	101	105	90.80	6.11
Effect of Gemfibrozil (17.14 mg/kg/day)							
18	84	64	55	96	95	78.80	7.40
20	65	43	46	78	80	62.40	6.94
22	56	33	39	48	60	47.20	4.52

Table-6
Effect of Cholesterol + Butter Rich Diet on the Serum Triglycerides mg/100 ml
in Group 8 Treated 4 (T.4) Rabbits

Weeks	6A	6B	6C	6D	6E	Mean	± SE	
0	45	25	26	26	40	32.40	3.76	
4	85	40	45	28	65	52.60	9.00	
8	98	70	70	35	82	71.00	9.27	
12	125	85	93	37	97	87.40	12.78	
16	140	120	150	40	130	116.00	17.57	
Effect of fish oil (71.43 mg/kg/day) + lovastatin (0.57 mg/kg/day)								P
18	110	60	104	30	105	81.80	14.12	NS
20	91	43	93	28	77	66.40	11.74	NS
22	50	38	52	22	46	41.60	4.88	<0.01

Table-7
Effect of Cholesterol + Butter Rich Diet on the Serum Triglycerides mg/100 ml
in Group 7 Treated 5 (T.5) Rabbits

Weeks	7A	7B	7C	7D	7E	Mean	± SE	
0	60	83	86	17	54	56.00	9.74	
4	170	90	75	25	65	85.00	21.31	
8	172	101	89	40	76	95.60	10.38	
12	186	109	110	77	98	116.00	16.53	
16	220	130	140	100	125	143.00	18.20	
Effect of fish oil (71.43 mg/kg/day) + gemfibrozil (17.14 mg/kg/day)								P
18	200	110	110	80	100	120.00	18.55	NS
20	53	95	34	30	82	58.80	11.54	<0.05
22	45	28	21	16	36	29.20	4.64	<0.01

Table-8
Effect of Cholesterol + Butter Rich Diet on the Serum Triglycerides mg/100 ml
in Group 8 Treated 6 (T.6) Rabbits

Weeks	8A	8B	8C	8D	8E	Mean	± SE	
0	43	24	10	18	12	21.40	5.30	
4	58	26	73	29	23	41.80	8.95	
8	69	31	128	42	29	59.80	16.53	
12	98	88	195	59	41	96.20	23.80	
16	121	123	218	76	52	118.00	25.42	
Effect of Lovastatin (0.57 mg/kg/day) + gemfibrozil (17.14 mg/kg/day)								P
18	69	74	170	44	39	79.20	21.20	NS
20	49	55	140	27	27	59.60	18.68	NB
22	18	22	23	14	19	18.80	1.53	<0.05

Table-9
Effect of Cholesterol + Butter Rich Diet on the Serum Triglycerides mg/100 ml in Group 9 Treated 7 (T.7) Rabbits

Weeks	9A	98	9C	9D	9E	Mean	± SE	P
0	70	54	56	57	61	59.60	2.54	
4	136	72	78	70	05	84.20	11.73	
8	179	96	89	73	73	102.00	17.68	
12	210	120	94	86	95	121.00	20.54	
16	250	140	130	120	160	160.00	20.98	
Effect of fish oil (71.43 mg/kg/day) + Lovastatin (0.57 mg/kg/day) + gemfibrozil (17.14 mg/kg/day)								P
18	190	130	104	36	80	108.00	22.97	NS
20	169	60	59	33	49	74.00	21.68	<0.05
22	31	50	46	30	33	38.00	3.72	<0.01

CONCLUSION

It can be concluded that a combination of gemfirozil and lovastatin showed the best profile in lowering triglycerides, whereas in case of fish oil and lovastatin or fish oil and gemfibrozil the effect is not so pronounced as the mode of action is not synergistic but rather it can exhibit a mechanism of antagonism in case of fish oil and lovastatin.

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